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Binomial Regression

Review Learning Objectives

1. The domain (input) of the binomial regression link function $g(p) = \log\left(\frac{p}{1-p}\right)$ is the interval $[0, 1]$. 1 / 1 point

- ☒ True
☐ False
☐ Correct

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2. The "probit" link function $h(\eta) = \Phi^{-1}(\eta)$ where Φ^{-1} is the inverse of the standard normal cdf, p is the probability of success from the binomial response, and η is the linear predictor. 3 / 3 points

- ☒ True
☐ False
☐ Correct

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3. The likelihood function for binomial regression is the joint pmf of the response, but interpreted as a function of the parameters of the model (with the response data fixed). 3 / 3 points

- ☒ True
☐ False
☐ Correct

4. The likelihood function and the log-likelihood function: 3 / 3 points

- ☐ both have the same maximum value.
☒ are both maximized at the same input/parameter value.
☐ Correct

5. Let event E have probability p of occurrence. Then the odds in favor of E is defined as: $\frac{p}{1-p}$. 3 / 3 points

- ☐ True
☒ False
☐ Correct

6. Suppose that the probability of contracting a virus v is $p = 0.1$. What are the odds of contracting v ? 3 / 3 points

- ☐ 1/11
☒ 1/9
☐ 11
☐ 9
☐ Correct

7. Consider data on the survival of patients who had undergone surgery for breast cancer. The data consists of a response (survival status after five years) and two predictors (the age of the patient at the time of the operation, and the number of cancerous axillary nodes detected). 4 / 4 points

1. x_1 : Age of patient in years at time of operation (**predictor**)
2. x_2 : Number of cancerous axillary nodes detected (**predictor**)
3. Y_1 : Survival status (**response**): 0 = the patient survived 5 years or longer; 1 = the patient died within 5 year

Suppose that a logistic regression model, with standardized predictors, correctly fits the data:

$$\eta = \beta_0 + \beta_1 z_1 + \beta_2 z_2 = \log\left(\frac{p}{1-p}\right),$$

where p is the probability of a patient surviving 5 years or longer, and

$$z_j = \frac{x_j - \min(x_j)}{\max(x_j) - \min(x_j)} \text{ for } j = 1, 2.$$

Which of the following are correct?

- ☐ β_0 represents the mean log odds of surviving 5 years or longer for a person of minimum age and with no cancerous axillary nodes detected.
☒ β_0 represents the mean log odds of surviving 5 years or longer for a person of (sample) mean age, and with the (sample) mean number of cancerous axillary nodes detected.

Correct

☐ For a fixed number of cancerous axillary nodes detected, a one year increase in age increases the log-odds of survival beyond 5 years by β_1 , on average.

☒ For a fixed number of cancerous axillary nodes detected, a one standard deviation increase in age increases the odds of survival beyond 5 years by a multiplicative factor of e^{β_1} , on average.

Correct

☒ For a fixed number of cancerous axillary nodes detected, a one standard deviation increase in age increases the log-odds of survival beyond 5 years by β_1 , on average.

Correct

8. Consider a logistic regression model that uses data to estimate the probability that a client will default on a monthly credit card payment (defaulting on a payment means that the client fails to pay their bill by the deadline for the month in question). 4 / 4 points

1. x_1 : credit limit in dollars (**predictor**)
2. x_2 : dollar amount of the bill statement one month prior (**predictor**)
3. x_3 : dollar amount of the bill statement for two months prior (**predictor**)
4. x_4 : dollar amount of the payment one month prior (**predictor**)
5. x_5 : dollar amount of the payment two months prior (**predictor**)
6. Y_1 : default status (**response**): 0 = the client did not default on the payment for the month in question; 1 = the client did default on the payment for the month in question.

Suppose that a logistic regression model correctly fits the data:

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 = \log\left(\frac{p}{1-p}\right),$$

where p is the probability of default.

☐ e^{β_1} is the average odds of default for a one-thousand dollar increase in credit limit, adjusting for the prior two months' bill statement and payment amounts.

☒ e^{β_5} represents the mean odds of defaulting on a payment for a person with a \$0 credit limit, a \$0 bill statement for the last two months, and \$0 in payments for the last two months.

Correct

☐ $\beta_1/1000$ is the average log-odds of default for a one-thousand dollar increase in credit limit, adjusting for the prior two months' bill statement and payment amounts.

☒ β_5 represents the mean log-odds of defaulting on a payment for a person with a \$0 credit limit, a \$0 bill statement for the last two months, and \$0 in payments for the last two months.

Correct